Calculus II

Worksheet

Force on a Spring = k * Displacement
Work = Force * Distance
Hydrostatic Force = Density * Depth * Area
Mass = Density * Area
Moment = Distance * Mass
Center of Mass = $(\overline{x}, \overline{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m}\right)$

1. A spring has a natural length of 5 ft. A force of 4 lb stretches this spring 1/4 ft.
   (a) Find the work done in stretching this spring from its natural length to 7 ft.
   (b) find the work done in stretching this spring from 8 ft to 9 ft.
2. How much work is done in emptying a cylindrical tank of radius 5 m and height 10 m by pumping the the water through a pipe extending 4 m above the tank. (Hint: Water weighs 9800 N/m^3, or 62.4 lb/ft^3.)
3. A rectangular gate built into a dam is 4 ft high and 8 ft wide. The top of the gate is 2 ft below the surface of the water. What is the force on the gate?
4. What would be the force if the gate were the upper half of disc of radius 4 (and the top of the gate is 2 ft below the surface of the water)?
5. Find the mass, moments, and center of mass of a the triangular lamina bounded by the x- and y- axes and $y = 4 - 2x$. (As always, assume the density of this lamina to be 1.)